

Q1

```
import numpy as np
arr = np.array([10, 20, 30, 40])
print(arr)
```

Q2

```
import numpy as np
arr = np.array([[1, 2, 3], [4, 5, 6]])
print(arr)
```

Q3

```
import numpy as np
arr = np.arange(1, 9).reshape(2, 2, 2)
print(arr)
```

Q4

```
import numpy as np
zeros = np.zeros(5, dtype=int)
ones = np.ones((3, 4), dtype=int)
print(zeros)
print(ones)
```

Q5

```
import numpy as np
arr = np.arange(0, 21, 4)
print(arr)
```

Q6

```
import numpy as np
arr = np.random.randint(1, 101, size=(2, 3, 4))
print(arr)
```

Q7

```
import numpy as np
arr = np.eye(4)
print(arr)
```

Q8

```
import numpy as np
arr = np.linspace(0, 1, 6)
```

```
print(arr)
```

Q9

```
import numpy as np
arr = np.array([[1, 2, 3], [4, 5, 6]])
print(arr)
```

Q10

```
import numpy as np
arr = np.array([[[1, 2], [3, 4]], [[5, 6], [7, 8]]])
print("ndim:", arr.ndim)
print("shape:", arr.shape)
print("dtype:", arr.dtype)
print("size:", arr.size)
```

Q11

```
import numpy as np
arr = np.zeros((4, 5))
print(arr.shape)
```

Q12

```
import numpy as np
arr = np.array([1, 2, 3])
print(arr.dtype)
arr = arr.astype(float)
print(arr.dtype)
```

Q13

```
import numpy as np
arr = np.array([1, 2, 3], dtype=np.int8)
print(arr)
print(arr.dtype)
```

Q14

```
import numpy as np
arr = np.zeros((3, 4, 2))
print("ndim:", arr.ndim)
print("size:", arr.size)
```

Q15

```
import numpy as np
a = np.array([1, 2, 3], dtype=np.int32)
b = np.array([1.5, 2.5, 3.5], dtype=np.float64)
c = a + b
print("a:", a.dtype)
print("b:", b.dtype)
print("result:", c.dtype)
```

Q16

```
import numpy as np
arr = np.arange(10)
print(arr[2])
print(arr[-1])
```

Q17

```
import numpy as np
arr = np.array([[1, 2, 3], [4, 5, 6]])
print(arr[0, 1])
```

Q18

```
import numpy as np
arr = np.array([[1, 2, 3], [4, 5, 6]])
print("row:", arr[0, :])
print("column:", arr[:, 1])
```

Q19

```
import numpy as np
arr = np.arange(1, 9).reshape(2, 2, 2)
print(arr[1, 0, 1])
```

Q20

```
import numpy as np
arr = np.array([[1, 2], [3, 4]])
arr[0, 1] = 99
print(arr)
```

Q21

```
import numpy as np
arr = np.array([[1, 2, 3], [4, 5, 6]])
```

```
print("last row:", arr[-1, :])  
print("last column:", arr[:, -1])
```

Q22

```
import numpy as np  
arr = np.arange(10)  
print(arr[2:7])
```

Q23

```
import numpy as np  
import numpy as np  
arr = np.arange(10)  
print(arr[:2])
```

Q24

```
import numpy as np  
arr = np.arange(10)  
print(arr[:-1])
```

Q25

```
import numpy as np  
arr = np.arange(1, 13).reshape(4, 3)  
print(arr[:2, :3])
```

Q26

```
import numpy as np  
arr = np.arange(1, 13).reshape(3, 4)  
print(arr[:, ::2])
```

Q27

```
import numpy as np  
arr = np.arange(1, 13).reshape(4, 3)  
print(arr[:-1, :])
```

Q28

```
import numpy as np  
arr = np.arange(1, 17).reshape(4, 4)  
arr[1:3, 1:3] = [[50, 60], [70, 80]]  
print(arr)
```

Q29

```
import numpy as np
arr = np.array([1, 2, 3, 4])
view = arr[1:3]
view[0] = 99
print("original:", arr)
print("view:", view)
```

Q30

```
import numpy as np
arr = np.array([1, 2, 3, 4])
copy_arr = arr.copy()
copy_arr[0] = 99
print("original:", arr)
print("copy:", copy_arr)
```

Q31

```
import numpy as np
arr = np.array([1, 2, 3])
view = arr[1:]
copy_arr = arr.copy()
print(view.base is arr)
print(copy_arr.base is arr)
```

Q32

```
import numpy as np
arr = np.array([1, 2, 3, 4])
part = arr[1:3]
part[:] = 99
print(arr)
```

Q33

```
import numpy as np
arr = np.array([1, 2, 3])
copy_arr = arr.copy()
view = arr.view()
copy_arr[0] = 10
print("after copy:", arr)
```

```
view[1] = 20
print("after view:", arr)
```

Q34

```
import numpy as np
arr = np.array([10, 20, 30, 40])
print("max:", arr.max())
print("min:", arr.min())
print("mean:", arr.mean())
```

Q35

```
import numpy as np
arr = np.array([[1, 2, 3], [4, 5, 6]])
print("axis=0:", arr.min(axis=0))
print("axis=1:", arr.min(axis=1))
```

Q36

```
import numpy as np
arr = np.array([[1, 2, 3], [4, 5, 6]])
print("column-wise:", arr.mean(axis=0))
print("row-wise:", arr.mean(axis=1))
```

Q37

```
import numpy as np
arr = np.array([1, 2, 3, 4, 5])
print("std:", arr.std())
print("var:", arr.var())
```

Q38

```
import numpy as np
arr = np.array([10, 50, 30, 90, 20])
print(arr.argmax())
```

Q39

```
import numpy as np
arr = np.array([[1, 2, 3], [4, 5, 6]])
print("row-wise:", arr.sum(axis=1))
print("column-wise:", arr.sum(axis=0))
print("overall:", arr.sum())
```

Q40

```
import numpy as np
arr = np.random.randint(1, 101, 10)
print("array:", arr)
print("max:", arr.max())
print("min:", arr.min())
print("mean:", arr.mean())
print("std:", arr.std())
print("var:", arr.var())
```

Q41

```
import numpy as np
arr = np.arange(1, 13)
print(arr.reshape(3, 4))
```

Q42

```
import numpy as np
arr = np.arange(1, 13)
print(arr.reshape(2, 3, 2))
```

Q43

```
import numpy as np
arr = np.arange(1, 13)
print(arr.reshape(3, -1))
```

Q44

```
import numpy as np
arr = np.arange(1, 13).reshape(3, 4)
print(arr.flatten())
print(arr.ravel())
```

Q45

```
import numpy as np
arr = np.arange(1, 7)
try:
    print(arr.reshape(4, 2))
except ValueError as e:
    print("Error:", e)
```

Q46

```
import numpy as np
arr = np.array([10, 20, 30])
for i, value in enumerate(arr):
    print(i, value)
```

Q47

```
import numpy as np
arr = np.array([[1, 2], [3, 4], [5, 6]])
for row in arr:
    print(row)
```

Q48

```
import numpy as np
arr = np.array([[1, 2], [3, 4]])
for row in arr:
    for value in row:
        print(value)
```

Q49

```
import numpy as np
arr = np.arange(1, 9).reshape(2, 2, 2)
for value in np.nditer(arr):
    print(value)
```

Q50

```
import numpy as np
arr = np.arange(1, 11)
for value in np.nditer(arr[::2]):
    print(value)
```

Q51

```
import numpy as np
a = np.array([1, 2, 3])
b = np.array([4, 5, 6])
print(np.concatenate((a, b)))
```

Q52

```
import numpy as np
```

```
a = np.array([[1, 2], [3, 4]])
b = np.array([[5, 6], [7, 8]])
print("vstack:")
print(np.vstack((a, b)))
print("hstack:")
print(np.hstack((a, b)))
```

Q53

```
import numpy as np
arr = np.arange(1, 10)
parts = np.array_split(arr, 3)
for p in parts:
    print(p)
```

Q54

```
import numpy as np
arr = np.arange(1, 13).reshape(4, 3)
parts = np.vsplit(arr, 2)
for p in parts:
    print(p)
```

Q55

```
import numpy as np
a = np.array([1, 2])
b = np.array([3, 4, 5])
c = np.array([6, 7, 8, 9])
joined = np.concatenate((a, b, c))
parts = np.array_split(joined, [2, 5])
print(joined)
for p in parts:
    print(p)
```

Q56

```
import numpy as np
arr = np.random.randint(1, 100, 10)
print(np.sort(arr))
```

Q57

```
import numpy as np
arr = np.array([[3, 1, 2], [6, 4, 5]])
print(np.sort(arr, axis=1))
```

Q58

```
import numpy as np
arr = np.array([5, 2, 9, 1, 7])
print(np.sort(arr)[::-1])
```

Q59

```
import numpy as np
arr = np.array([3, 1, 2])
arr.sort()
print("in-place:", arr)
arr2 = np.array([3, 1, 2])
sorted_arr = np.sort(arr2)
print("original:", arr2)
print("new array:", sorted_arr)
```

PART B — PANDAS

Q60

```
import pandas as pd
s = pd.Series([10, 20, 30], index=['a', 'b', 'c'])
print(s)
```

Q61

```
import pandas as pd
s = pd.Series({'x': 1, 'y': 2, 'z': 3})
print(s)
```

Q62

```
import pandas as pd
df = pd.DataFrame({
    'Name': ['A', 'B', 'C', 'D'],
    'Age': [20, 21, 22, 23],
    'City': ['Jaipur', 'Delhi', 'Mumbai', 'Pune']
})
print(df)
```

Q63

```
import pandas as pd
data = [['A', 20], ['B', 21], ['C', 22]]
df = pd.DataFrame(data, columns=['Name', 'Age'])
print(df)
```

Q64

```
import pandas as pd
df = pd.DataFrame({'Name': ['A', 'B', 'C']},
index=['r1', 'r2', 'r3'])
print(df)
```

Q65

```
import pandas as pd
s = pd.Series([10, 20, 30], name='Marks')
df = s.to_frame()
print(df)
```

Q66

```
import numpy as np
import pandas as pd
data = np.random.randint(1, 101, size=(5, 3))
df = pd.DataFrame(data, columns=['A', 'B', 'C'])
print(df)
```

Q67

```
import pandas as pd
df = pd.read_csv('data.csv')
print(df.shape)
```

Q68

```
import pandas as pd
df = pd.read_csv('data.csv', sep=';')
print(df)
```

Q69

```
import pandas as pd
df = pd.read_csv('data.csv', nrows=5)
print(df)
```

Q70

```
import pandas as pd
df = pd.read_csv('data.csv', index_col='ID')
print(df)
```

Q71

```
import pandas as pd
df = pd.read_csv('data.csv', header=None,
names=['Name', 'Age', 'City'])
print(df)
```

Q72

```
import pandas as pd
df = pd.read_csv('data.csv', usecols=['Name', 'Age'])
print(df)
```

Q73

```
print(df['Name'])
print(df[['Name', 'Age']])
```

Q74

```
print(df.loc[0])
print(df.iloc[0])
```

Q75

```
print(df.at[0, 'Name'])
print(df.loc[0, 'Name'])
```

Q76

```
df = df.rename(columns={'Name': 'Student_Name'})
print(df)
```

Q77

```
df.columns = ['Name', 'Age', 'City']
print(df)
```

Q78

```
df.index = ['r1', 'r2', 'r3']
print(df)
```

Q79

```
df = df.reset_index(drop=True)
```

```
print(df)
```

Q80

```
df['Total'] = df['Math'] + df['Science']
```

```
print(df)
```

Q81

```
print(df.head(10))
```

Q82

```
print(df.tail(3))
```

Q83

```
print(df.info())
```

```
print(df.isna().sum())
```

Q84

```
print(df.describe())
```

Q85

```
print(df.describe(include='object'))
```

Q86

```
import pandas as pd
```

```
df = pd.DataFrame({'A': [1, None, 3], 'B': [4, 5, None]})
```

```
print(df.isna())
```

Q87

```
print(df.isnull().sum())
```

Q88

```
df = df.dropna()
```

```
print(df)
```

Q89

```
df = df.dropna(how='all')
```

```
print(df)
```

Q90

```
df = df.dropna(axis=1)
```

```
print(df)
```

Q91

```
df.fillna(0, inplace=True)
```

```
print(df)
```

Q92

```
df['Age'].fillna(df['Age'].mean(), inplace=True)  
print(df)
```

Q93

```
print(df[df.duplicated()])
```

Q94

```
df.drop_duplicates(inplace=True)  
print(df)
```

Q95

```
df.drop_duplicates(subset=['Name'], inplace=True)  
print(df)
```

Q96

```
print(df['Age'].mean())
```

Q97

```
print(df['Age'].median())
```

Q98

```
print(df['City'].mode())
```

Q99

```
df['Age'].fillna(df['Age'].median(), inplace=True)  
print(df)
```

Q100

```
print("Mean:", df['Age'].mean())  
print("Median:", df['Age'].median())  
print("Mode:", df['Age'].mode().tolist())
```